Pre-Deployment Bias Review Protocol

For machine-learning classification systems operating on regulated health data · Sai Kuruguntla · IWU AIML-501

Reviewer's value statement

I will treat my own certainty as the least-tested control in the system.

Every control I put in place gets reviewed and documented. My judgment about *which* controls matter usually does not. That asymmetry is where bias lives in security work, so this protocol exists to review the reviewer.

When to run this

Before any machine-learning system is granted read access to a store containing protected health information, and again on any change to its training data, prompt, model version, or threshold. The reference case is an LLM classifier labeling documents on two axes — PHI / Non-PHI and Internal / External — so access controls can be applied automatically.

The five gates

Stage	Question the reviewer must answer	Evidence required	Fails if
1. Data selection	Was the training and test data drawn from the whole population, or only from the parts I already knew about?	A population inventory produced *before* sampling, plus a written record of what was excluded and why.	Exclusions are undocumented or discovered after the fact.
2. Labeling	Whose definition of PHI became ground truth, and would a second qualified person agree with it?	Two independent labelers on the held-out set, with measured agreement and an adjudication log.	A single labeler defined ground truth, or disagreements were resolved silently.
3. Design and thresholds	Does the system treat a missed PHI file as equal in cost to an over-flagged file?	A stated error-cost position, a threshold tuned for recall on the PHI class, and a defined low-confidence path to human review.	The model is forced to choose a label on every item with no abstain option.
4. Testing and evaluation	Does the reported metric hide a segment where the model performs badly?	Precision and recall broken out by document type, language, and owning team — not one aggregate figure — plus a hand-read sample of false negatives.	Only aggregate accuracy is reported, or no one has read a false negative.
5. Interpretation	Will a downstream reader treat a generated label as a verified fact?	Confidence surfaced alongside every label, an explicit uncertain state, and a named channel for reporting wrong labels after launch.	Labels are presented without confidence, or there is no way to contest one.

Rollout conditions

A bias review that ends at approval is incomplete, because the biases that matter most surface after real users meet the system. Four conditions apply to any deployment that clears the gates above:

- **Pilot narrowly first.** One team, one document set, with results reviewed before the scope widens.
- **State the limits in writing.** What the classifier is known to get wrong, said plainly, to the people who will rely on it.
- **Keep the reporting channel open and answered.** A team that has been oversold on an AI system stops reporting its errors, and at that point the bias becomes invisible and permanent.
- **Re-run the gates on drift.** New document types, a new model version, or a changed threshold all re-open Gate 4.

Adapted from an AIML-501 discussion assignment on navigating human bias in AI/ML change leadership. The five stages follow the recognized human entry points for bias in AI systems: data selection, labeling, algorithm design, testing and evaluation, and interpretation of results. Drafted and refined with Claude (Anthropic); the review positions, error-cost reasoning, and pass/fail conditions are my own.